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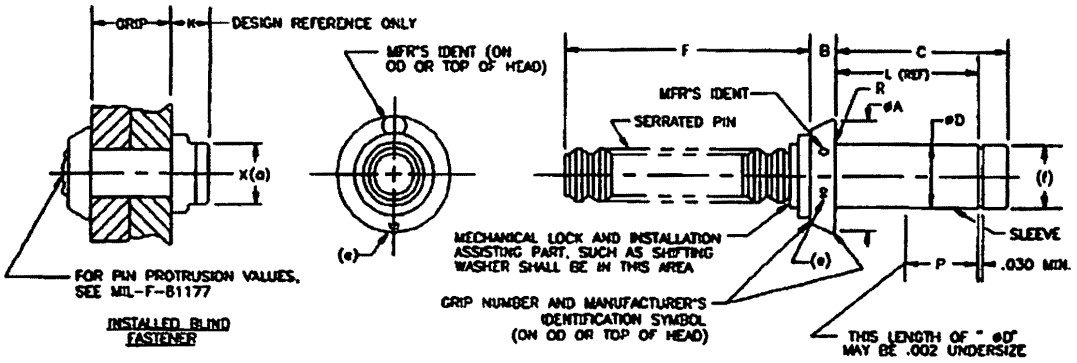


TABLE I. FASTENER DIMENSIONS AND STRENGTHS

SIZE DASH NUMBER (b)				øA	B	øD	F	K	P	R	X	INSTALLED FASTENER		
DOUBLE ACTION	SINGLE ACTION		NOM. SIZE									MINIMUM SINGLE SHEAR LBS.	MINIMUM TENSILE LBS.	HOLE LIMITS
(c)	(d)	(d)					MIN.	MAX.		RAD MAX.	(e) MIN.			
~~-05~~	S05	U05	5/32	.272 .250	.070 .062	.164 .162	.844	.202	.215	.010	.195	2340	1350	.164 .167
~~-06~~	S06	U06	3/16	.332 .305	.135 .125	.199 .197	.875	.231	.250	.015	.238	3450	2100	.199 .202
-	-08	U08	1/4	.432 .400	.140 .130	.260 .258	1.000	.279	.305	.018	.315	5900	3650	.260 .263
-	-10	U10	5/16	.522 .480	.141 .131	.312 .310	1.218	.319	.350	.021	.373	8500	5200	.312 .315
-	-12	U12	3/8	.627 .580	.205 .195	.374 .372	1.562	.364	.405	.025	.448	12200	7500	.374 .377
-	-14	U14	7/16	.727 .675	.207 .197	.437 .435	1.562	.419	.460	.030	.522	16700	10150	.437 .441
-	-16	U16	1/2	.832 .770	.270 .260	.499 .498	1.562	.540	.598	.035	.598	21800	13500	.500 .504

(a) MAXIMUM DIAMETER OF BLIND HEAD (x) SHALL NOT EXCEED MAXIMUM DIAMETER OF MANUFACTURED HEAD (A).

(b) SIZES "05" AND "06" WITH "-" ARE INSTALLED WITH A DOUBLE ACTION TOOL. ALL OTHER SIZES ARE INSTALLED WITH A SINGLE ACTION TOOL. SIZES "U05" AND "U06" MAY BE INSTALLED WITH A DOUBLE OR SINGLE ACTION TOOL. SEE MIL-F-81177, APPENDIX A FOR TOOL SELECTION CHART.

(c) ~~-05, -06, -595, -596, -08, -10, -12, -14 AND -16~~ ARE INACTIVE FOR NEW DESIGN AFTER 1 JAN 91. USE U05, U06, S05 AND S06.

(d) LETTER "U" SIGNIFIES SINGLE ACTION FASTENER WITHOUT FRANGIBLE TYPE OF LOCKING COLLAR OR LOCKING COLLAR CAP. "U" AND "S" CODE FASTENERS IN SIZES "05" AND "06" ARE COLOR IDENTIFIED GREEN ON LOCKING COLLAR, SHIFTING WASHER, OR SERRATED PIN. "S" CODE FASTENERS MAY BE SUPPLIED WITH A LOCKING COLLAR CAP.

(e) SINGLE DIGIT MARKING IS PERMISSIBLE FOR GRIP DASH NUMBERS LESS THAN 10.

(f) DIAMETER OF SERRATED PIN HEAD SHALL NOT EXCEED "øD" MAX. AND MAY BE .002 BELOW "øD" MIN. IN AREA "P" ONLY.

(T) DENOTES CHANGE(S)

INCH-POUND

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY-AV NAVY-AS
AIR FORCE-99 DLA-IS
REVIEW: MC, 82
USER:
PROJECT NUMBER: 5320-0616

MILITARY SPECIFICATION SHEET

TITLE
FASTENER, BLIND, HIGH STRENGTH, PULL TYPE, POSITIVE MECHANICAL LOCK, PROTRUDING HEAD, ALLOY STEEL, 112 K.S.I. Fsu

SPECIFICATION SHEET NUMBER
MS90354 28 FEB 94 REV T

SUPERSEDING
MS90354 R 13 AUG 1993

AMSC N/A FSC 5320

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TABLE II. GRIP NUMBER TABULATION

05/32 (-05)				03/16 (-06)				01/4 (-08)				05/16 (-10)			
GRIP DASH NO.	GRIP RANGE MIN.	GRIP RANGE MAX.	C ⓤ MAX.	GRIP DASH NO.	GRIP RANGE MIN.	GRIP RANGE MAX.	C ⓤ MAX.	L REF.	GRIP DASH NO.	GRIP RANGE MIN.	GRIP RANGE MAX.	C ⓤ MAX.	L REF.	GRIP DASH NO.	GRIP RANGE MIN.	GRIP RANGE MAX.	C ⓤ MAX.	L REF.
-0501	.031	.095	.372	-0601	.094	.157	.479	.303	-0801	.094	.157	.553	.344	-1001	.094	.157	.618	.381
-0502	.094	.157	.434	-0602	.156	.220	.541	.365	-0802	.156	.220	.616	.406	-1002	.156	.220	.680	.443
-0503	.156	.220	.496	-0603	.219	.282	.604	.428	-0803	.219	.282	.678	.469	-1003	.219	.282	.743	.506
-0504	.219	.282	.559	-0604	.281	.345	.666	.490	-0804	.281	.345	.740	.531	-1004	.281	.345	.805	.568
-0505	.281	.345	.622	-0605	.344	.407	.729	.553	-0805	.344	.407	.803	.594	-1005	.344	.407	.868	.631
-0506	.344	.407	.684	-0606	.406	.470	.791	.615	-0806	.406	.470	.866	.656	-1006	.406	.470	.930	.693
-0507	.406	.470	.746	-0607	.469	.532	.854	.678	-0807	.469	.532	.928	.719	-1007	.469	.532	.993	.756
-0508	.469	.532	.809	-0608					-0808					-1008				
-0509	.531	.595	.872	-0609	.531	.595	.916	.740	-0809	.531	.595	.990	.781	-1009	.531	.595	1.055	.818
-0510	.594	.657	.934	-0610	.594	.657	.979	.803	-0810	.594	.657	1.053	.844	-1010	.594	.657	1.118	.881
-0511	.656	.720	.996	-0611	.656	.720	1.041	.865	-0811	.656	.720	1.116	.906	-1011	.656	.720	1.180	.943
-0512	.719	.782	1.059	-0612	.719	.782	1.104	.928	-0812	.719	.782	1.178	.969	-1012	.719	.782	1.243	1.006
-0513	.781	.845	1.122	-0613	.781	.845	1.166	.990	-0813	.781	.845	1.240	1.031	-1013	.781	.845	1.305	1.068
-0514	.844	.907	1.184	-0614	.844	.907	1.229	1.053	-0814	.844	.907	1.303	1.094	-1014	.844	.907	1.368	1.131
-0515	.906	.970	1.246	-0615	.906	.970	1.291	1.115	-0815	.906	.970	1.366	1.156	-1015	.906	.970	1.430	1.193
-0516	.969	1.032	1.309	-0616	.969	1.032	1.354	1.178	-0816	.969	1.032	1.428	1.219	-1016	.969	1.032	1.493	1.256
				-0617	1.031	1.095	1.416	1.240	-0817	1.031	1.095	1.490	1.281	-1017	1.031	1.095	1.555	1.318
				-0618	1.094	1.157	1.479	1.303	-0818	1.094	1.157	1.553	1.344	-1018	1.094	1.157	1.618	1.381
				-0619	1.156	1.220	1.541	1.365	-0819	1.156	1.220	1.616	1.406	-1019	1.156	1.220	1.680	1.443
				-0620	1.219	1.282	1.604	1.428	-0820	1.219	1.282	1.678	1.469	-1020	1.219	1.282	1.743	1.506
				-0621	1.281	1.345	1.666	1.490	-0821	1.281	1.345	1.740	1.531	-1021	1.281	1.345	1.805	1.568
				-0622	1.344	1.407	1.729	1.553	-0822	1.344	1.407	1.803	1.594	-1022	1.344	1.407	1.868	1.631
				-0623	1.406	1.470	1.791	1.615	-0823	1.406	1.470	1.866	1.656	-1023	1.406	1.470	1.930	1.693
				-0624	1.469	1.532	1.854	1.678	-0824	1.469	1.532	1.928	1.719	-1024	1.469	1.532	1.993	1.756
				-0625	1.531	1.595	1.916	1.740	-0825	1.531	1.595	1.990	1.781	-1025	1.531	1.595	2.055	1.818
				-0626	1.594	1.657	1.979	1.803	-0826	1.594	1.657	2.053	1.843	-1026	1.594	1.657	2.118	1.881
				-0627	1.656	1.720	2.041	1.865	-0827	1.656	1.720	2.116	1.906	-1027	1.656	1.720	2.180	1.943
				-0628	1.719	1.782	2.104	1.928	-0828	1.719	1.782	2.178	1.969	-1028	1.719	1.782	2.243	2.006
									-0829	1.781	1.845	2.240	2.031	-1029	1.781	1.845	2.305	2.068
									-0830	1.844	1.907	2.303	2.094	-1030	1.844	1.907	2.368	2.131
														-1031	1.906	1.970	2.430	2.193
														-1032	1.969	2.032	2.493	2.256

PREPARING ACTIVITY: NAVY-AS
CUSTODIANS: ARMY-AV NAVY-AS
AIR FORCE-99 DLA-IS
REVIEW MI, 82
USER:

MILITARY SPECIFICATION SHEET

TITLE
FASTENER, BLIND, HIGH STRENGTH, PULL TYPE, POSITIVE MECHANICAL LOCK, PROTRUDING HEAD, ALLOY STEEL, 112 K.S.I. Fsu

SPECIFICATION SHEET NUMBER
MS90354 28 FEB 94 REV T
SUPERSEDING MS90354 R 13 AUG 1993

AMSC- N/A FSC 5320

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TABLE II. GRIP NUMBER TABULATION (CONTINUED)

GRIP DASH NO.	ø3/8 (-12) GRIP RANGE MIN.	ø3/8 (-12) GRIP RANGE MAX.	ø3/8 (-12) C MAX.	ø3/8 (-12) L REF.	ø7/16 (-14) GRIP DASH NO.	ø7/16 (-14) GRIP RANGE MIN.	ø7/16 (-14) GRIP RANGE MAX.	ø7/16 (-14) C MAX.	ø7/16 (-14) L REF.	ø1/2 (-16) GRIP DASH NO.	ø1/2 (-16) GRIP RANGE MIN.	ø1/2 (-16) GRIP RANGE MAX.	ø1/2 (-16) C MAX.	ø1/2 (-16) L REF.
-1202	.294	.157	.595	.423	-1404	.219	.282	.901	.599	-1605	.281	.345	1.084	.753
-1203	.156	.220	.757	.485	-1405	.281	.345	.964	.662	-1606	.344	.407	1.147	.815
-1204	.219	.282	.820	.548	-1406	.344	.407	1.026	.724	-1606	.406	.470	1.209	.878
-1205	.281	.345	.883	.610	-1407	.406	.470	1.089	.787	-1607	.469	.532	1.272	.940
-1206	.344	.407	.945	.673	-1408	.469	.532	1.151	.849	-1608				
-1207	.406	.470	1.007	.735										
-1208	.469	.532	1.070	.798										
-1209	.531	.595	1.133	.860	-1409	.531	.595	1.214	.912	-1609	.531	.595	1.334	1.003
-1210	.594	.657	1.195	.923	-1410	.594	.657	1.276	.974	-1610	.594	.657	1.397	1.065
-1211	.656	.720	1.257	.985	-1411	.656	.720	1.339	1.037	-1611	.656	.720	1.459	1.128
-1212	.719	.782	1.320	1.048	-1412	.719	.782	1.401	1.099	-1612	.719	.782	1.522	1.190
-1213	.781	.845	1.383	1.110	-1413	.781	.845	1.464	1.162	-1613	.781	.845	1.584	1.253
-1214	.844	.907	1.445	1.173	-1414	.844	.907	1.526	1.224	-1614	.844	.907	1.647	1.315
-1215	.905	.970	1.507	1.235	-1415	.906	.970	1.589	1.287	-1615	.906	.970	1.709	1.378
-1216	.969	1.032	1.570	1.298	-1416	.969	1.032	1.651	1.349	-1616	.969	1.032	1.772	1.440
-1217	1.031	1.095	1.633	1.360	-1417	1.031	1.095	1.714	1.412	-1617	1.031	1.095	1.834	1.503
-1218	1.094	1.157	1.695	1.423	-1418	1.094	1.157	1.776	1.474	-1618	1.094	1.157	1.897	1.565
-1219	1.156	1.220	1.757	1.485	-1419	1.156	1.220	1.839	1.537	-1619	1.156	1.220	1.959	1.628
-1220	1.219	1.282	1.820	1.548	-1420	1.219	1.282	1.901	1.599	-1620	1.219	1.282	2.022	1.690
-1221	1.281	1.345	1.883	1.610	-1421	1.281	1.345	1.964	1.662	-1621	1.281	1.345	2.084	1.753
-1222	1.344	1.407	1.945	1.673	-1422	1.344	1.407	2.026	1.724	-1622	1.344	1.407	2.147	1.815
-1223	1.406	1.470	2.007	1.735	-1423	1.406	1.470	2.089	1.787	-1623	1.406	1.470	2.209	1.878
-1224	1.469	1.532	2.070	1.798	-1424	1.469	1.532	2.151	1.849	-1624	1.469	1.532	2.272	1.940
-1225	1.531	1.595	2.133	1.860	-1425	1.531	1.595	2.214	1.912	-1625	1.531	1.595	2.334	2.003
-1226	1.594	1.657	2.195	1.923	-1426	1.594	1.657	2.276	1.974	-1626	1.594	1.657	2.397	2.065
-1227	1.656	1.720	2.257	1.985	-1427	1.656	1.720	2.339	2.037	-1627	1.656	1.720	2.459	2.128
-1228	1.719	1.782	2.320	2.048	-1428	1.719	1.782	2.401	2.099	-1628	1.719	1.782	2.522	2.190
-1229	1.781	1.845	2.383	2.110	-1429	1.781	1.845	2.464	2.162	-1629	1.781	1.845	2.584	2.253
-1230	1.844	1.907	2.445	2.173	-1430	1.844	1.907	2.526	2.224	-1630	1.844	1.907	2.647	2.315
-1231	1.906	1.970	2.507	2.235	-1431	1.906	1.970	2.589	2.287	-1631	1.906	1.970	2.709	2.378
-1232	1.968	2.032	2.570	2.298	-1432	1.968	2.032	2.651	2.349	-1632	1.969	2.032	2.772	2.440
-1233	2.031	2.095	2.633	2.360	-1433	2.031	2.095	2.714	2.412	-1633	2.031	2.095	2.834	2.503
-1234	2.094	2.157	2.695	2.423	-1434	2.094	2.157	2.776	2.474	-1634	2.094	2.157	2.897	2.565
-1235	2.156	2.220	2.757	2.485	-1435	2.156	2.220	2.839	2.537	-1635	2.156	2.220	2.959	2.628
-1236	2.219	2.282	2.820	2.548	-1436	2.219	2.282	2.901	2.599	-1636	2.219	2.282	3.022	2.690
										-1637	2.281	2.345	3.084	2.753
										-1638	2.344	2.407	3.147	2.815
										-1639	2.406	2.470	3.208	2.878
										-1640	2.469	2.532	3.272	2.940

THE SOLICITATION: MIL-F-81177

PREPARING ACTIVITY: NAVY - AS
CUSTODIANS: ARMY - AV NAVY - AS
AIR FORCE - 99 DLA - IS
REVIEW: MI, 82
USER:
PROJECT NUMBER 5320-0816

MILITARY SPECIFICATION SHEET

TITLE
FASTENER, BLIND, HIGH STRENGTH, PULL TYPE, POSITIVE MECHANICAL LOCK, PROTRUDING HEAD, ALLOY STEEL, 112 KSI

SPECIFICATION SHEET NUMBER
MS90354 28 FEB 94 REV T

SUPERSEDING
MS90354 1 AUG 1993

AMSC - N/A FSC 5320

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EQUIREMENT:

1. MATERIAL

   SLEEVE AND SERRATED PIN - ALLOY STEEL PER ASTM-A-331

   MECHANICAL LOCK: CARBON STEEL OR ALLOY STEEL CONFORMING TO THE CHEMICAL COMPOSITION OF 1010, 1020, 1144, 4037, OR 4130 OF FEDERAL STANDARD 66 OR A286 CRES CONFORMING TO THE CHEMICAL LIMITS OF AMS 5737, OR TYPE 316 CRES CONFORMING TO THE CHEMICAL REQUIREMENTS OF AMS5690.

2. HARDNESS

   SLEEVE: KNOOP (1000 GRAM LOAD) 326-466

   SERRATED PIN: ROCKWELL "C" 48 TO 53

   MECHANICAL LOCK: NOT APPLICABLE

3. PROTECTIVE TREATMENT

   SLEEVE: CADMIUM PLATE PER QQ-P-416, TYPE II, CLASS 2 OR ALUMINUM COATING PER MIL-C-83488.

   SERRATED PIN: CADMIUM PLATE PER QQ-P-416, TYPE I, CLASS 3 WITH HYDROGEN EMBRITTLEMENT RELIEF PER MIL-F-81177 OR ALUMINUM COATING PER MIL-C-83488.

   MECHANICAL LOCK: CADMIUM PLATE PER QQ-P-416, TYPE II, CLASS 2 (CARBON AND ALLOY STEEL). PASSIVATE PER QQ-P-35 (A286 AND TYPE 386 CRES) OR ALUMINUM COATING PER MIL-C-83488, AT MANUFACTURER'S OPTION.

4. INSPECTION: DUE TO INSTALLATION LOADS BEING HIGHER THAN PERFORMANCE LOAD REQUIREMENTS, MAGNETIC PARTICLE INSPECTION IS NOT REQUIRED.

5. LUBRICATION: NONE, EXCEPT MANUFACTURER'S OPTION TO LUBRICATE LOCK MECHANISM WITH LOX COMPATIBLE DRY FILM LUBRICANTION.

6. EXAMPLE OF PART NUMBER:

MS 90354 - 0808 D - 1/4 DIA. NOM SIZE, .156 MIN. GRIP, INSTALLED WITH A SINGLE ACTION TOOL.

- "D" DENOTES ALUMINUM COATING; NO CODE DENOTES CADMIUM PLATING.
- GRIP RANGE (SEE TABLE II).
- SIZE DASH NUMBER (SEE TABLE I).
- "-" DENOTES FASTENERS INSTALLED WITH A DOUBLE ACTION TOOL, FOR SIZES 05 AND 06.
- "-" DENOTES FASTENERS INSTALLED WITH A SINGLE ACTION TOOL, FOR SIZES 08, 10, 12, 14, AND 16.
- "S" DENOTES FASTENERS INSTALLED WITH A SINGLE ACTION TOOL, FOR SIZES 05 AND 06.
- "U" DENOTES SINGLE ACTION FASTENERS INSTALLED WITH A SINGLE ACTION TOOL, FOR ALL SIZES (SEE SHEET 1, FOOTNOTE (d)).
- MS NUMBER

OTES:

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2. IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS DOCUMENT AND THE REFERENCES CITED HEREIN, THE TEXT OF THIS DOCUMENT SHALL TAKE PRECEDENCE.
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